

Electricity : Circuits and their Components

CHAPTER 01

Learning Outcomes...

- Electric Charges and Electric Current
- Electric Circuit and Symbol of its Components
- Electric Potential and Potential Difference
- Heating effect of Electric Current
- Electric Fuse
- Magnetic Effect of Electric Current
- Electromagnet

Introduction 13/4/26

As soon as we press the switch of a torch ON, it starts glowing. For the torch bulb to glow, there must be something flowing in the wires. This 'something' must have energy, otherwise it cannot produce heat and light energy in the torch bulb. We call this 'something' as electric current. An electric current has electrical energy. This electrical energy gets converted into light and heat energy when the current flows through an electric bulb. The electric current flows from the end of the cell marked '+' (called the positive terminal) through the metal wires, to the end marked '-' (called the negative terminal). The arrangement where an electric current flows in a conductor when it is connected to a source of electric current is known as an electric circuit.

In this chapter, we will study about electric components and devices which are represented by symbols in a circuit diagram, heating effect of current and magnetic effects of current. In magnetic effects of current we will deal with magnetic effects by current flowing in a wire.

Torch, Simple Electrical Circuit and their Symbol

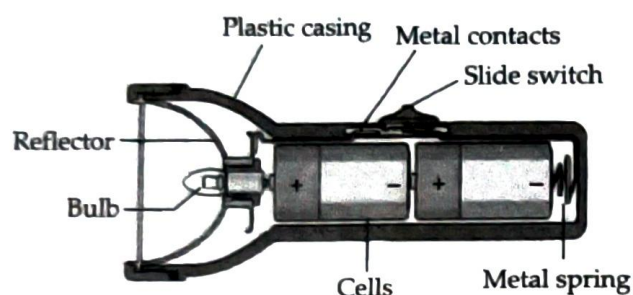
A Torchlight

A torch, or flashlight, is a portable electric light that works by converting stored electrical energy from a battery into light. An electric torch works by completing an electric circuit when the switch is turned on, allowing electricity to flow from the batteries to a bulb or LED. This current makes the bulb's filament (or the LED) glow, producing light. A reflector and lens then focus and direct the light, creating a beam.

Construction: A torch contains the following basic parts:

- **Battery:** This is the power source. It stores chemical energy and turns it into electrical energy. It has a positive (+) and a negative (-) end.
- **Switch:** This component is used to close and open the electric circuit. When you turn the switch on, it completes the path, allowing current to flow. When you turn it off, it breaks the path.
- **Bulb (or LED):** This is the component that produces light.
 - **Incandescent bulb:** Contains a thin filament, usually made of tungsten, that glows when heated by the electric current.

- **LED (Light-Emitting Diode):** A more energy-efficient alternative that uses a semiconductor to produce light.
- **Wire and Spring:** Wire is a link between the bulb and the switch. Spring is a link between the cells and the switch. When the switch is pushed to the 'on' position, the circuit is complete and the bulb glows. When the switch is pushed to the 'off' position, the circuit is incomplete (broken) the bulb does not glow.
- **Reflector and lens:** A reflector is placed behind the bulb to reflect and direct the light forward, while a lens focuses this light into a beam.



ILLUSTRATIONS

1 What is the use of reflector in torch?

Ans.: A reflector in a torch uses to collect and focus the light from the bulb into a powerful

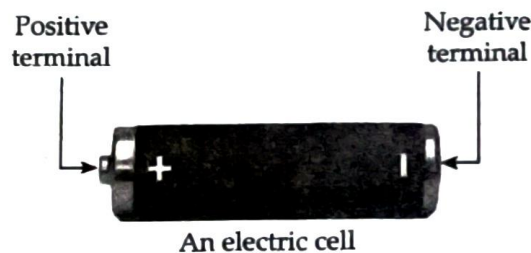
2 What is the use of spring in torch?

Ans.: Spring is a link between the cells and the switch.

A Simple Electrical Circuits

Electric Cell

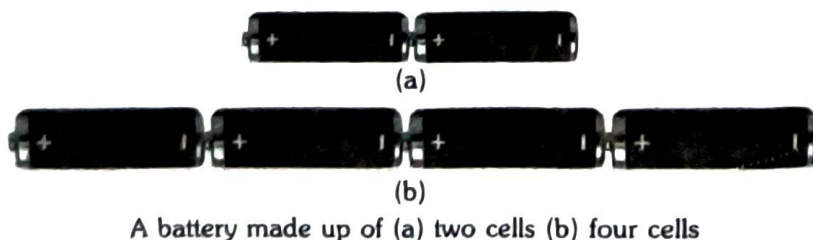
An electric cell is a portable source of electrical energy. All electric cells have two terminals; one is called a positive terminal (+) while the other is negative terminal (-). The metal cap serves as the positive terminal of the electrical cell, while the metal disc acts as the negative terminal.



Battery

A battery consists of two or more electric cells connected to each other. There is usually a thick wire or a metal strip connecting the positive terminal of one cell to the negative terminal of the next. To place the cells correctly in a battery compartment, '+' and symbols are usually printed there on the dry cell or storage battery.

The cells in a torch are placed in a specific order to form a battery. When cells are placed with the positive terminal of one cell connected to the negative terminal of the next, the lamp glows. If the cells are not connected in series correctly (positive to negative), the lamp does not glow.



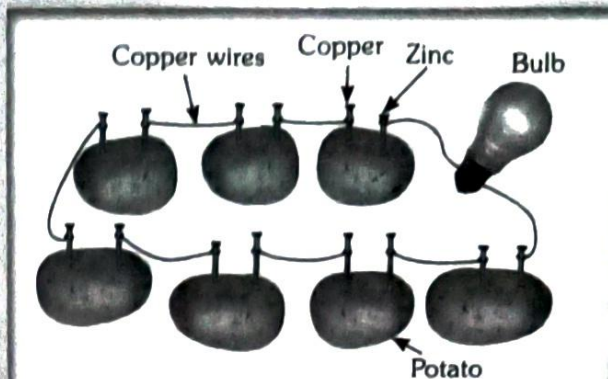
A battery made up of (a) two cells (b) four cells

ACTIVITY CORNER



To make your own battery to produce electricity.

- » Take seven medium sized potatoes. In each potato, insert a zinc strip on one end and a copper strip on the other end.
- » Connect the zinc strip of one potato to the copper strip of the other potato using copper wires.
- » Connect the electric bulb as shown in the figure.
- » You will see that as soon as you connect the bulb, it glows.
- » The bulb glows because the whole arrangement of potatoes acts as a battery.



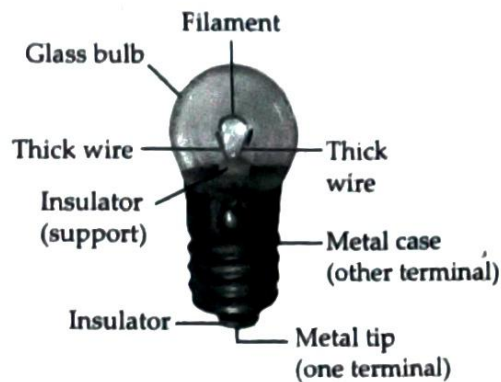
Electric Lamp

Incandescent Lamp

An incandescent lamp is a light bulb that uses a thin, glowing wire called a filament to produce light. When electricity flows through the filament, it heats up to a very high temperature and glows. **The filament is protected from oxidation by a glass bulb that is either filled with an inert gas or has its air removed.**

Components of an incandescent lamp :

- **Glass Bulb:** A glass enclosure that protects the filament.
- **Tungsten Filament:** A thin wire that heats up and glows when electricity passes through it. Tungsten is used because it can withstand very high temperatures.
- **Contact Wires:** Wires that connect the filament to the bulb's electrical contacts at the base.
- **Support Wires:** Wires that hold the filament in place.
- **Glass Mount:** A glass support that holds the filament and contact wires.
- **Screw Cap:** The metal base of the bulb that fits into a light socket, providing both mechanical support and electrical contact.



ACTIVITY CORNER



Let us observe

- » For this activity, you will require a torchlight with an incandescent lamp (or light bulb). Many old torchlights still use such lamps. With your teacher's help, confirm that your torchlight uses an incandescent

lamp. Take the torch and examine its lamp.

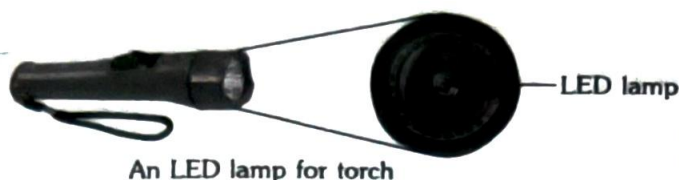
- » What do you see? Do you notice a thin wire fixed in the middle of the glass bulb? Yes, there is a thin wire (filament) fixed in the middle of the glass bulb. When the torch is switched ON, the thin wire inside the glass bulb glows. After inspecting the lamp, we find that the filament is attached to two thicker wires that support it. One wire connects to the metal case at the lamp's base, and the other connects to the metal tip at the center of the base, forming the two terminals which are fixed so they don't touch each other as shown in figure.



LED Lamp

An LED lamp is a type of light that uses a small electronic component called an LED (Light Emitting Diode) to produce light. When electricity passes through the LED, it lights up. Most of the torches that we use today have an LED lamp instead of an incandescent lamp.

Unlike incandescent lamps, LEDs do not have filaments but have two terminals, one is positive (longer wire), and the other is negative (shorter wire). Thus, a torch may use one or more LEDs and have different shapes.



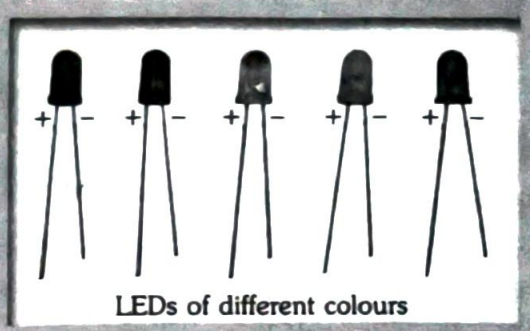
ACTIVITY CORNER



Let us observe

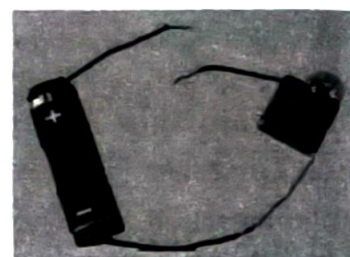
► Take an LED of any colour and observe. Notice the length of two wires attached to the LED.

► Do you see any filament inside it. No, there is no filament inside the LED's. Also, one wire (the positive terminal) is longer than the other (the negative terminal).



Making an electric lamp glow using an electric cell or battery.

The electric current flows through the lamp when the circuit is complete, causing the lamp to glow. The direction of the electric current flows from the positive terminal to the negative terminal of the electric cell in the circuit.



For incandescent lamps, it does not matter

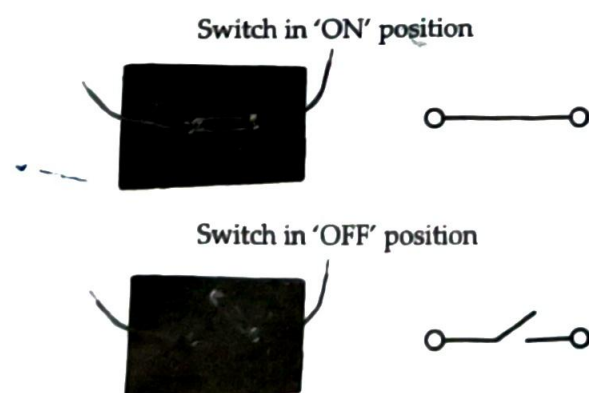
which terminal (positive or negative) is connected to which terminal of electric cell. The lamp will glow as long as the circuit is complete and current flows through its filament.

Electric Switch

A switch is a simple device that either completes or breaks a circuit. The switches used for lights and the other devices at home work the same way, though they are designed differently. To understand 'ON' and 'OFF' position of switch, let us take a simple arrangement as shown in figure, containing pins, safety pin and two connecting wires. This arrangement is the representation of switch in circuit.

'ON' Position: When the safety pin touches both drawing pins, it closes the gap, completing the circuit and allowing current to flow. The current flows from the positive terminal to the negative terminal of the electric cell, making the lamp glow.

'OFF' Position: When the safety pin does not touch the second drawing pin, there is a gap in the circuit, preventing current from flowing. This means the lamp does not glow. The circuit is considered open in this position.



ILLUSTRATIONS

3 How does an electric cell produce electricity?

Ans.: The electric cell produces electricity due to chemical reactions that occurs inside it.

4 Name a safe source of electricity for performing science experiments in school laboratory.

Ans.: Electric cells and batteries.

5 Why is the inert gas filled in electric bulb?

Ans.: Electric bulbs are usually filled with chemically inactive gases (inert gases) like nitrogen and argon. These gases are filled in an electric bulb to prevent the tungsten filament from burning out due to oxidation.

6 What is the full name of LED?

Ans.: "Light Emitting Diode"

7 What is the melting point of tungsten?

Ans.: The melting point of tungsten is 3422°C .

Symbols of Electric Components

In electric circuits, we have to show various electrical components such as a cell, a battery, connecting wires, an open switch, a closed switch, and an electric bulb (or lamp), etc. Now, to draw the electric circuits by making the actual sketches of the various electrical components is a difficult job and takes a lot of time. So, the scientists have devised some symbols for electrical components which are easy to draw. They are called electrical symbols or circuit symbols. The common electrical symbols for electrical components which are used in drawing circuit diagrams are given below:

Component	Description	Symbol
Wire 	Represented by a straight line, provides a medium for the flow of current in a circuit, usually made of copper and coated with plastic.	
Cell 	In the symbol, the longer line represents the positive terminal and the shorter one represents the negative terminal of the cell, simpler source of electric current.	
Battery 	A combination of two or more cells.	
Switch 	Used to open or close the circuit.	 Close/ON or  Open/OFF
Electric Lamp 	An electrical appliance containing filament that glows when the current flows through it.	



Electric Charges

- ▶ Electric charge of a material body or particle is the property due to which it produces and experiences electrical and magnetic effects.
- ▶ Charges are of two types : positive charge and negative charge.
- ▶ Proton is a positively charged particle present in the nucleus of an atom. Electrons are negatively charged particle which revolve around the nucleus.

Properties of electric charges :

- ▶ Like charges repel and unlike charges attract each other.
- ▶ Electric charges can neither be created nor be destroyed in an isolated system. This means electric charge is conserved.

COMPETITION WINDOW

- ▶ Electric charges are quantised i.e., the total charge of the body is the integral multiple of charge of electrons.

$$\therefore q = ne ; \text{ where } n \text{ is an integer and } e \neq 1.6 \times 10^{-19} \text{ C.}$$

- ▶ The SI unit of charge is coulomb (C).

Electric Current

- ▶ The rate of flow of electric charge through a conductor is called the electric current and it is denoted by I .

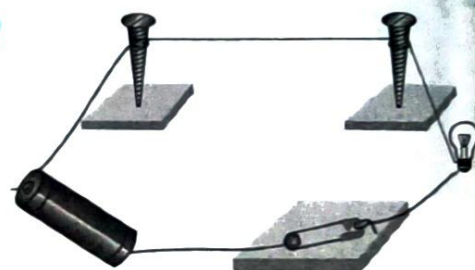
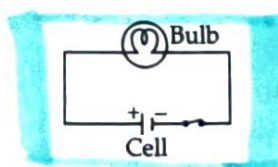
$$I = \frac{Q}{t}$$

- ▶ The SI unit of electric current is ampere which is denoted as A.

Electric Circuit

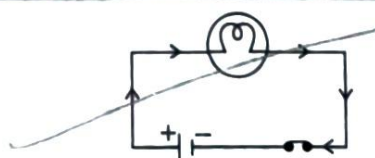
The closed continuous path through which an electric current flows is called an electric circuit. An electric circuit can be represented by a simple and clear circuit diagram. The electric circuits can be drawn using electric components shown in figure.

Using the symbols, the electric circuit can be drawn as follows :

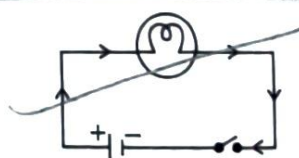


When the switch is in 'ON' condition, the circuit is closed, the electric current flows from the positive terminal of the battery to negative terminal and instantly the bulb glows.

When the switch is in 'OFF' condition, the circuit is open, no electric current flows along this broken or incomplete path through any part of the circuit and the bulb does not glow.



Closed circuit



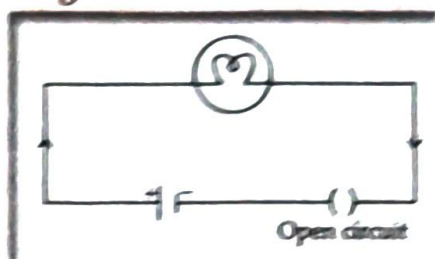
Open circuit

Usually copper wires are used for making connections in a circuit as they are good conductors of electric current. But they are covered with plastic, which is a bad conductor of electricity, so that the user does not get a shock. Many devices such as torches, toys, TV remote controls use batteries to work.

ILLUSTRATIONS

8 What do you understand by an open circuit?

Ans.: An open circuit occurs when there is a break in the circuit which stops the flow of current. Given figure shows an open circuit. Breaks in the circuit can be due to many causes, such as a loose lamp holder or a missing wire in the circuit.



9 Match the following in column I to those in column II.

Column I	Column II
i. Conductor	a. a collection of cells.
ii. Circuit	b. used to open or close a circuit.
iii. Insulator	c. generates electricity.

iv. Cells

v. Battery

vi. Switch

d. allows electricity to flow through it.

e. complete path for the flow of electricity.

f. substances that do not allow electricity to flow through it.

Ans:

(i) Conductor	→ d. allows electricity to flow through it.
(ii) Circuit	→ e. complete path for the flow of electricity
(iii) Insulator	→ f. substances that do not allow electricity to flow through it.
(iv) Cells	→ c. generates electricity
(v) Battery	→ a. a collection of cells.
(vi) Switch	→ b. used to open or close a circuit

Electric Potential and Potential Difference

» Electric potential at a point is the work done in moving a positive unit charge from infinity to that point.

» Potential difference between two points is the amount of work done in moving a unit positive charge from one point to the other.

$$\text{Electric potential} = \frac{\text{work}}{\text{charge}} \text{ or } V = \frac{W}{Q}$$

» The SI unit of electric potential is volt (V).

Ohm's Law

» The current flowing through a metallic conductor is directly proportional to the

COMPETITION WINDOW

potential difference applied across the ends of the metallic conductor, provided the physical condition remain the same i.e., $I \propto V$ or $V \propto I$

$\Rightarrow V = IR$; where R is the proportionality constant called the resistance of the conductor.

» Resistance is the property of a conductor to resist the flow of charge through it.

» SI unit of resistance is ohm and is denoted by Ω .

» According to Ohm's law, $R = \frac{V}{I}$



DO YOU KNOW?

An 'ammeter' is the circuit element which is used to measure the current flowing in an electric circuit. Similarly, a 'voltmeter' is used to measure the electric potential difference between two points in an electric circuit.

Effects of Electric Current

When electric current flows through a conductor, it can produce many effects. We make use of these effects in various real life applications. When the current flows through a circuit, we see various effects of current depending on the components being used. We have already observed the lighting effects of the electric current in a glowing bulb. Let us study about the heating and magnetic effects of electric current.

Heating Effect of Electric Current

When an electric current is passed through a wire, the wire becomes hot and produces heat. This is called the heating effect of current. For example, the filament of bulb is the part which gives light, when a bulb is connected in a circuit and switched on, its filament becomes so hot that it gives out light. When bulb glows for a long time, the current also flows through it for long time, the longer the bulb glows the hotter it becomes. Therefore, it can be concluded that :

- (i) The electric current flowing through a conductor produces heating effect.
- (ii) Longer the current flowing through the circuit, higher the heating effect produced by it.

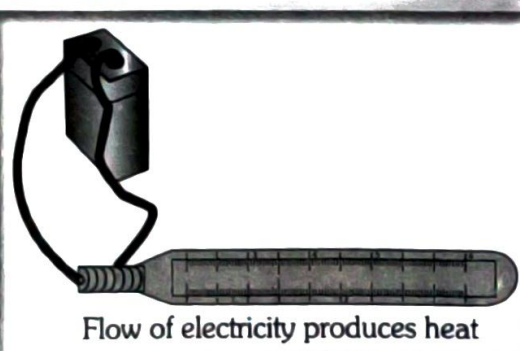
ACTIVITY CORNER



To show that electric current produces heat.

- Take a 4.5 volt battery, long insulated (copper) wire and a thermometer. Wind a copper wire around the bulb of the thermometer so that the spirals do not touch themselves and the ends of the wire should be fairly long. Connect the ends of the wire to the terminals of a 4.5 volt battery.

- After few minutes, you will observe that the temperature rises on the thermometer, this is because the electric current that flows through the wire develops heat.



Some appliances such as electric heater, electric immersion rod, electric toaster, electric iron, etc., also use this property of heating due to electric current. They have wound coils of wire known as 'heating element' through which current flows resulting in its heating due to resistance offered by the wire.

The amount of heat generated in a wire depends on the material, length and thickness of the wire. Thus for different requirements, the wires of different materials and different lengths and thickness are used.

An electric bulb produces heat while producing light. This leads to wastage of electricity. Use of compact fluorescent lamps (CFL's), LED's can reduce this wastage to a large extent. However, before buying any electrical appliances, one should look for the ISI mark of the Bureau of Indian standards. This ensures that the appliance is safe and wastage is minimum.

Tungsten and nichrome get too hot on passing current through them. It makes them useful for making filaments of electric bulbs and other heating devices (heaters, irons, geysers, toasters etc.).

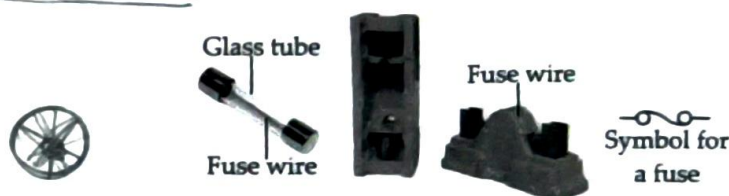


Electric bulbs have a filament made of tungsten while the filaments of electric iron, electric heaters, geysers, toasters are made of nichrome.

Tungsten and nichrome have the ability to withstand very high temperature without melting.

Electric Fuse

Electric fuse is a safety device used in electrical circuits. It is used to prevent fires and protect electrical devices from damage.



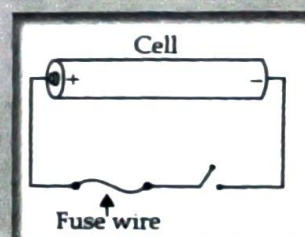
A fuse is a small piece of wire of an alloy of adequately low melting point (usually 63% tin and 37% lead) connected to an electric circuit. There is a maximum limit of the current which can safely flow through a circuit. If the current exceeds the safe limit, then the fuse wire gets heated up and melts. This breaks the circuit and the appliance is saved from getting burnt.

ACTIVITY CORNER



To show the working of a fuse.

- ▶ Take a battery of four cells, a thin strand of steel wool (steel wool commonly used for cleaning utensils), two connecting wires. Connect the strand of steel wool to the circuit containing battery using connecting wires. The steel wool will serve as a fuse wire.
- ▶ Now, close the key of the circuit. Observe the strand of steel wool. The steel strand melts down and breaks the circuit. The current stops flowing in the circuit. This is how a fuse works.



Nowadays, a device called as **miniature circuit breaker (MCB)** is also used instead of fuses, in the household electric circuits. It is a switch that automatically stops the current in a circuit if the current in it exceeds the specified maximum limit.





Joule's Law of Heating

- ▶ If Q amount of charge flows through a conductor with potential difference V , then Work done is $W = QV$... (i)

But $I = Q/t \therefore I \times t$... (ii)

From equation (i) and (ii), $W = VIt$

Also by Ohm's law, $V = IR$

$$W = (IR)t = I^2Rt$$

This work is converted into heat energy,

$$H = I^2Rt$$

$\Rightarrow$ Heat energy, $H \propto I^2$ and $H \propto R$ and $H \propto t$

- ▶ **Overloading** : The condition in which too many electrical appliances of high power

COMPETITION WINDOW

rating are switched on at the same time, then the total current drawn by the appliances through the circuit may exceed its rated value. This is known as overloading.

- ▶ **Short circuit** : When two naked wires of electricity supply line touch each other, a large amount of current flow through the wires. This phenomenon is known as short circuiting.

Always, use proper fuses which have been specified for particular applications, carrying ISI mark. Never use just any wire or strip of metal in place of a fuse.

Magnetic Effect of Electric Current

When electric current passes through a wire carrying current, it behaves like a magnet. This is called the magnetic effect of current.

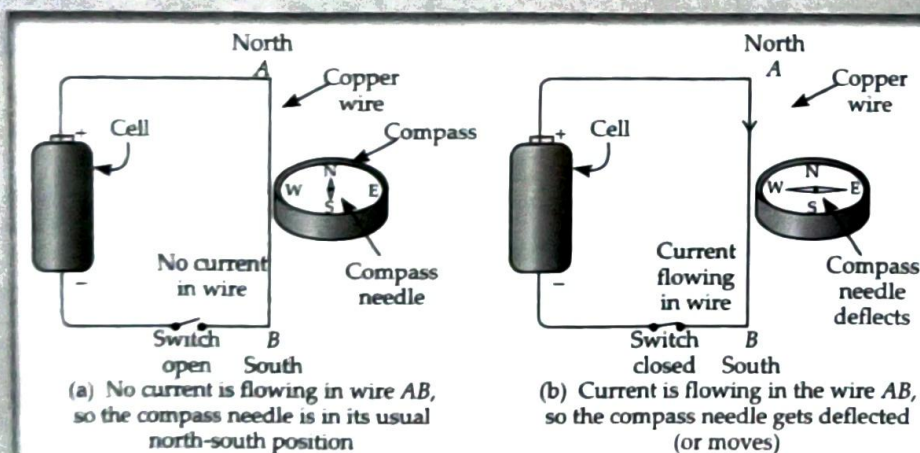
ACTIVITY CORNER



Understanding the magnetic effect of current.

- ▶ To understand the magnetic effect of current take a magnetic compass, place it on a table and allow it to come to rest. Remember, it will always come to rest, pointing in the north-south direction. Take a straight copper wire, about 10-12 inches long. Keep the wire parallel to the compass needle and connect its two ends to a cell and a switch to complete the circuit. Keep the

switch of the circuit 'on' to allow the current to flow through it, while carefully observing the direction of the compass needle. You will notice that as soon as the current starts flowing in the circuit, the needle shows a deflection (sudden change in its direction). Now, if you open the switch, you will see that the compass needle gets back to its original north-south direction. Repeat the experiment a few times, you will notice that the needle deflects when the current is switched on and stays in that position as long as the current flows in the wire. When the current is switched off, the needle comes back to its original north-south direction.



This experiment was first performed by Hans Christian Oersted in 1820 to show the magnetic effect of current. He concluded that when electric current passes through a wire, it behaves like a magnet.

DO YOU KNOW?

Extremely weak electric currents are produced in human body by the movement of charged particles called ions. Due to this a very weak magnetism is produced in the human body. This is the basis of the technique called Magnetic Resonance Imaging (MRI).

ILLUSTRATIONS

10 The heating element of a heater becomes red hot but not the connecting wire. Why?

Ans.: This is because the resistance of the heating wire in the element is high, whereas that of the connecting wire is extremely low.

11 What is the difference between a fuse wire and a heating element?

Ans.: A fuse wire is a safety device which prevents damages to electrical circuits and possible fires where as a heating element is used to heat up things. The fuse wire has low melting point, whereas the heating wire has a high melting point. Fuse wire is made up of tin-lead alloy, whereas heating wire is made of constantan.

12 Do you think an electromagnet can be used for separating plastic bags from a garbage heap? Explain.

Ans.: No, the plastic bags do not get attracted by the magnet, so they cannot be separated by an electromagnet. Plastic bags are not magnetic materials, only magnetic materials like iron can be attracted by the electromagnet.

13 An electrician is carrying out some repairs in your house. He wants to replace a fuse by a piece of wire. Would you agree? Give reasons for your response.

Ans.: No, we would not agree to allow to replace the fuse by a wire. Wires in the fuses are of specific melting points. So we should always use ISI marked fuses in our houses to prevent short circuits.

Electromagnet, Electrical Conductor and Insulator

Electromagnet

An electromagnet is based on the magnetic effect of current. It consists of a long length of insulated copper wire coiled around an iron rod or a bar. When current is passed through this coil, the iron rod behaves like a magnet. The magnetism of the iron rod disappears as soon as the current through the coil is switched off. Therefore, an electromagnet is a temporary magnet. Along with the current, the magnetism can also be switched 'OFF' or 'ON'.

The Factors on which the Strength of an Electromagnet depends

- The strength of an electromagnet depends directly upon
 - the amount of current flowing through the coil.
 - the number of turns in the coil.
- The strength of electromagnet also depends inversely upon the length of the electromagnet, which means lesser the length of electromagnet, more will be the strength of electromagnet.

Uses of Electromagnets

- In cranes to pick magnetic materials from a junkyard.
- In electric bells.
- In loudspeakers.
- By doctors to remove iron splinters from eyes.
- In electric motors.

DO YOU KNOW?

William Sturgeon is another scientist who experimented with electromagnets. He created one of the first practical electromagnets in 1824, which consisted of a horseshoe-shaped iron core wrapped with multiple turns of wire.

Electric Bell

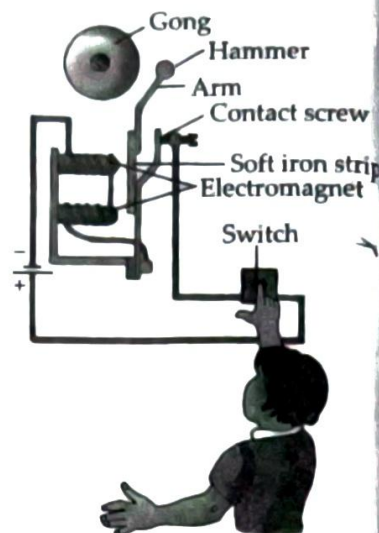
An electric bell works on the principle of magnetic effect of electric current.

It consists of a coil of wire wound on an iron piece. The coil acts as an electromagnet. An iron bar with a hammer at one end is kept close to the electromagnet. There is a contact screw near the iron strip. Initially when we switch on the circuit, current starts flowing in the circuit, and the iron piece wound by the coil gets magnetised. This electromagnet, then attracts the iron bar towards it.

When the current stops flowing, the iron bar comes back to its initial position as the electromagnet does not attract it any more. When it reaches the initial position, the circuit is complete again and the hammer strikes the gong again.

In the process, the hammer at the end of the strip strikes the gong of the bell to produce sound.

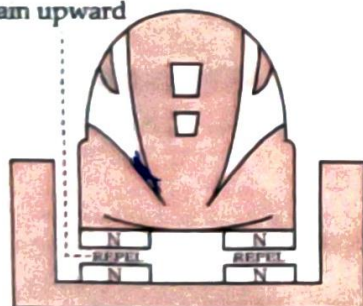
However, when the electromagnet pulls the iron strip, it also breaks the circuit. The current through the coil stops flowing. This process is repeated in quick succession. The hammer strikes the gong every time the circuit is completed. This is how the bell rings.



- Maglev trains work by using magnets to levitate the train above the tracks and propel it forward without any mechanical contact, which eliminates friction and allows for high speeds. Magnets are embedded in both the train and the guideway (track). The interaction between the magnets like poles repelling each other lifts the train, while the alternating attraction and repulsion of the magnetic fields propel it forward.

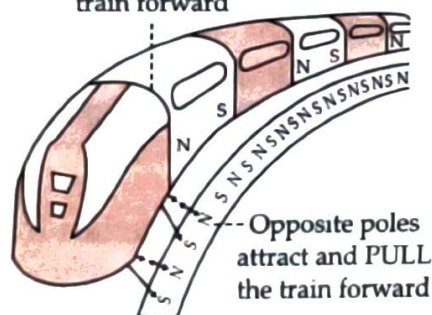
COMPETITION WINDOW

Like poles repel and PUSH the train upward



LEVITATION

Like poles repel and PUSH the train forward



PROPULSION

Electrical Conductors and Insulators

Conductors: Materials that allow electric current to pass through are called conductors. Metals are good conductors. Examples: silver, copper, and gold.

Insulators: Materials that do not allow electric current to pass through are called insulators or poor conductors of electricity. Examples: plastic, rubber, and ceramics.

Use of Conductors and Insulators

Metals like copper are used for making electrical wires because they are good conductors and relatively inexpensive. Insulators (like plastic, rubber, and ceramics) are used to cover wires because they stop electricity from passing through and keeps us safe from electric shocks.

The human body is a conductor of electricity. Electric current passing through the body can cause severe injury or death. Always handle electrical appliances with care. Avoid touching switches or plugs with wet hands, and never use electrical devices in wet areas or with damaged insulation or broken plugs.

DO YOU KNOW?

Silver is the most conductive of all metals, meaning it allows electric current to flow through it more freely than any other metal.

ACTIVITY CORNER

To identify conductors and insulators from a given set of materials (adult supervision required).

- » Fix the bulb and the cell on the cardboard sheet or wood, as shown in the figure.



- » Touch the two free ends of the wire to the two ends of various objects, such as a metal scale, an eraser, a glass, the tip of your pen, the tips of your pencil lead (the pointed portion and the flat portion at the back of the pencil), salt solution in a bowl and so on.
- » The bulb will glow if the object you touch with the free ends of the wire is a conductor, and the bulb will not glow if it is an insulator.
- » This way you will be able to classify various materials as conductor and insulator.

Application of Conductors and Insulators

Conductors are used in wires, switches, connectors, plugs, and sockets. Insulators are used to cover wires, plug tops, and switches to prevent electric shock.



COMPETITION WINDOW

- "A superconductor is defined as a substance that offers no resistance to the electric current when it becomes colder than a critical temperature."

Superconductors are different from ordinary conductors, such as copper. Unlike regular conductors whose resistance gradually reduces, the superconductor's resistance drops to zero below a fixed temperature, which is the critical temperature. At this temperature, a superconductor can conduct electricity with no resistance, which means no heat, sound, or other forms of energy would be discharged from the material when it reaches the "critical temperature" (T_c).



ILLUSTRATIONS

- 14 Describe an activity to show that you can make an electromagnet in your school laboratory.

Ans.: Take a long iron nail and wrap an insulated copper wire on it. Connect the end of the coil with a battery through a key. An

electromagnet is ready. When key is pressed current flows in the wire and the heating coil acts as a magnet.

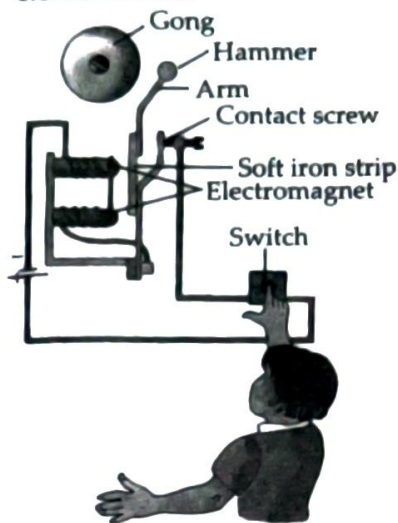
- 15 An electrician uses screwdrivers and pliers with plastic or rubber handles. Why?

Ans.: It is because rubber and plastic, both are insulators.

CONCEPT MAP

Electric Bell

An electric bell works on the principle of magnetic effect of electric current.



Electromagnet

- The magnet made by using electric current is called an electromagnet.
- Uses of electromagnets
 - in electric bells
 - in cranes to pick magnetic materials from junk yard.
 - in electric motors
 - in loud speakers



Magnetic Effect of Electric Current

- When electric current passes through a wire, the current - carrying wire behaves like a magnet. This is called magnetic effect of electric current.
- This is utilised in making electromagnets.

Electricity : Circuits and their Components

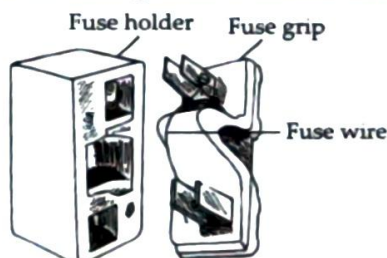
Heating Effect of Electric Current

- When an electric current passes through a high resistance wire, the electric energy gets converted into heat energy. This is known as heating effect of electric current.
- The electric heating appliances such as electric bulb, electric room heater, electric iron works on the principle of heating effect of current.



Electric Fuse

It is a safety device used in electric circuits. It works on the principle of heating effect of electric current.



Electric Circuit

A continuous conducting path between the two terminals of a cell or battery along which an electric current flows is known as electric circuit.

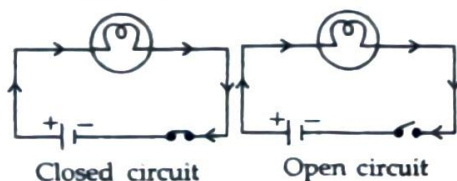


Symbols of Electrical Components

- Cell
- Battery
- Wire
- Electric Bulb
- Open key / switch
- Closed key / switch
- Fuse wire



Open and Closed Circuit



Solved Examples

1. What do you mean by electric current?

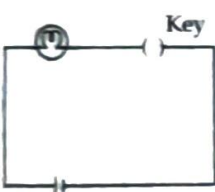
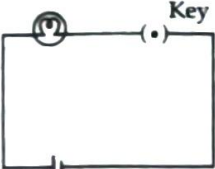
Ans.: The rate of flow of electric charges is known as electric current.

2. Define electric cell. What is the direction of electric current in a circuit with an electric cell?

Ans.: The simplest source of an electric current is an electric cell and it has a positive terminal and a negative terminal. In an electric circuit the current starts from positive terminal of the cell, passes through the equipment connected to it, and terminates at the negative terminal of the cell.

3. What is the difference between open circuit and closed circuit?

Ans.:

Open Circuit	Closed Circuit
1. A path which is open and current cannot flow through it.	1. A closed path in which current can flow.
2. 	2. 
3. Electrical devices cannot function in it.	3. Electrical devices can function.

4. What are the essential components of an electric circuit?

Ans.: The essential components of an electric circuit are electric cell, electric bulb, switch, battery and wires.

5. What is a fuse?

Ans.: Fuse is a safety device that is used in household circuits to protect the appliances. It prevents them from getting burnt out in case excess current flows through the circuit.

6. What is heating effect of the electric current?

Ans.: The wire or component or device gets hot when an electric current passes through them. This is known as heating effect of the electric current. *Definition*

7. How much work is done in moving a charge of 5 C across two points having potential difference 15 V?

Ans.: As charge $Q = 5 \text{ C}$

$$V = 15 \text{ V}$$

$$W = ?$$

$$\therefore W = VQ = 15 \text{ V} \times 5 \text{ C}$$

$$W = 75 \text{ J}$$

8. Name the factors on which the heat produced on an electric conductor depends.

Ans.: It depends on:

- the current for a given resistance.
- the resistance for a given current and
- the time for which the current flows through the resistor.

9. On what factors does the heat produced in a wire depends?

Ans.: Amount of heat produced in a wire depends on its material, length and thickness.

10. What do you mean by magnetic effect?

Ans.: When electric current passes through a wire, it behaves like a magnet. This is called magnetic effect of electricity.

11. How does the electric heater or an electric iron work?

Ans.: Electric heater or iron work on the basis of heating effect of electricity. When an electric current passes through the wires they become hot. Electric heater or iron contains a coil of wire. This coil of wire is called an element. When these appliances are switched on after connecting to

the electrical supply, their elements become red hot and release heat.

12. Distinguish between the terms 'overloading' and 'short circuiting'.

Ans.: Overloading : The condition in which too many electrical appliances of high power rating are switched on at the same time, then the total current drawn by the appliances through the circuit may exceed its rated value. This is known as overloading.

Short circuit : When two naked wires of electricity supply line touch each other, a large amount of current flow through the wires. This phenomenon is known as short circuiting.

13. Write the differences between a bar magnet and an electromagnet.

Ans.:

	Bar magnet		Electromagnet
1.	It is a permanent magnet.	1.	It is a temporary magnet.
2.	Its strength cannot be changed.	2.	Its strength can be changed.
3.	Its magnetic properties cannot be turned OFF.	3.	Its magnetic properties can be turned OFF.

14. What are the uses of electromagnets?

Ans.: Electromagnets are used in cars, speakers, electric motors, electric bells, telephone receivers, television, picture tubes etc.

15. What is an electric fuse made up of? How does it function to prevent a circuit from burning?

Ans.: An electric fuse is made up of an alloy (mixture of metals 63% tin and 37% lead), with low melting point. The heat produced due to the high current, quickly melts the fuse wire and break the circuit. Thus, stops the flow of current and damage is avoided.

16. Why do we cover plug pin holes which are within the reach of children with cellotape or a plastic cover when not in use?

(NCERT Exemplar)

Ans.: The child may put his/her fingers into the socket and he/she may get an electric shock which could be fatal. Plastic/cellotape are bad conductors of electricity and hence chances of electric shock gets reduced.

17. Can we use the same fuse in a geyser and a television set? Explain. (NCERT Exemplar)

Ans.: No, a geyser and a television set require different amount of current to operate properly. Therefore, the fuse used in these two appliances will be of different ratings.

18. Paheli does not have a night lamp in her room. She covered the bulb of her room with a towel in the night to get dim light. Has she taken the right step? Give one reason to justify your answer. (NCERT Exemplar)

Ans.: No, the towel may burn due to the heat produced by the bulb. Besides, it will also result in wastage of electric energy. She could have used a lamp shade.

19. Do all bulbs we use work on the principle of heating effects of electric current? Discuss.

Ans.: No, filament bulbs we use work on heating effect of electric current. In these bulbs, in addition to light, lot of unwanted heat is also generated and this results in the wastage of electricity. We have different types of bulbs, known as fluorescent tube lights which work on a different principle. Compact fluorescent lamps (CFL) also reduce wastage and can be fixed in the ordinary bulb holders.

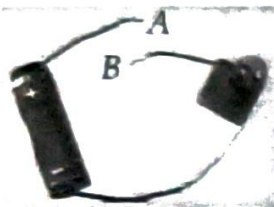
20. Boojho made an electromagnet by winding 50 turns of wire over an iron screw. Paheli also made an electromagnet by winding 100 turns over a similar iron screw. Which electromagnet will attract more pins? Give reason.

(NCERT Exemplar)

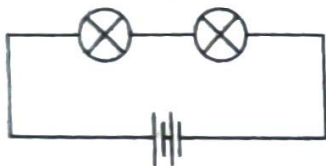
Ans.: Paheli's electromagnet will attract more pins as it has more number of turns of wire on it and thus a stronger electromagnet.

NCERT Section

- Choose the incorrect statement.
 - A switch is the source of electric current in a circuit.
 - A switch helps to complete or break the circuit.
 - A switch helps us to use electricity as per our requirement.
 - When the switch is in 'OFF' position, there is an air gap between its terminals
- Observe figure with which material connected between the ends A and B, the lamp will not glow?



- In figure, if the filament of one of the lamps is broken, will the other glow? Justify your answer.



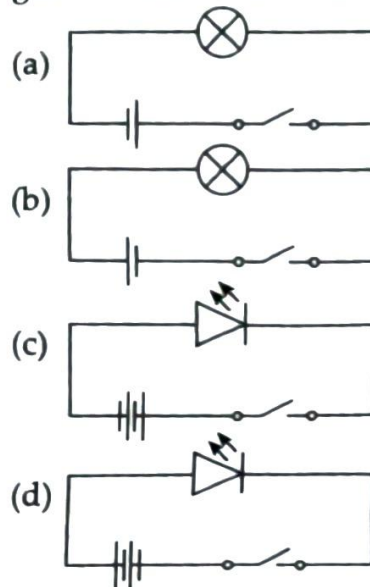
- A student forgot to remove the insulator covering from the connecting wires while making a circuit. If the lamp and the cell are working properly, will the lamp glow?
- Draw a circuit diagram for a simple torch using symbols for electric components.
- In figure : (i) If S_2 is in 'ON' position, S_1 is in 'OFF' position, which lamp(s) will glow?
(ii) If S_2 is in 'OFF' position, S_1 is in 'ON' position, which lamp(s) will glow?
(iii) If S_1 and S_2 both are in 'ON' position, which lamp(s) will glow?
(iv) If both S_1 and S_2 are in 'OFF' position, which lamp(s) will glow?

Vidyut has made the circuit as shown in figure. Even after closing the circuit, the lamp does not glow. What can be the possible reasons? List as many possible reasons as you can for this faulty operation. What will you

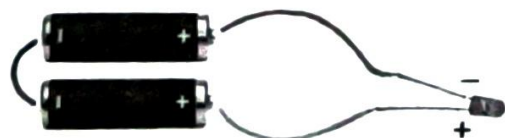
do to find out why the lamp did not glow?



- In figure in which case(s) the lamp will not glow when the switch is closed.



- Suppose the '+' and '-' symbols cannot be read on a battery. Suggest a method to identify the two terminals of this battery.
- You are given six cells marked A, B, C, D, E, and F. Some of these are working and some are not. Design an activity to identify which of them are working.
 - List the items that you require.
 - Write the procedure that you will follow.
 - With the items, carry out the activity to identify the cells that are working.
- An LED requires two cells in series to glow. Tanya made the circuit as shown in figure. Will the lamp glow? If not, draw the wires for correct connections.



Exercise

Multiple Choice Questions

LEVEL - 1

Torch, Simple Circuit and their Symbol

- The torch bulb is connected to the cell with the help of
(a) Single wire ☒ (b) Two wires
(c) Three wires (d) None of these.
(NCERT Page 24)
- What type of energy conversion happens in a cell when it powers a torch?
(a) Electrical energy to chemical energy.
☒ (b) Chemical energy to electrical energy.
(c) Light energy to electrical energy.
(d) Mechanical energy to electrical energy.
(Page 25)
- What is the source of electricity in a typical portable torch?
(a) The bulb
(b) The switch
☒ (c) The cells (batteries)
(d) The mains
(Page 25)
- The filament in a modern incandescent lamp is made of which material?
(a) Carbon ☒ (b) Tungsten
(c) Silver (d) Aluminium
(Page 26)
- What is the main disadvantage of incandescent lamps?
(a) They are high heat generation.
(b) They have a very short lifespan and low efficiency.
(c) They are difficult to dim.
☒ (d) Both (a) and (b).
(Page 26)
- Incandescent lamp is a
(a) Tube Light ☒ (b) Bulb
(c) Sodium Lamp (d) None of these
(Page 26)
- The thin wire inside a bulb which glows when electric current is passed through it is
(a) coil
(b) magnet
☒ (c) filament
(d) none of these. (Page 26) **R**
- An LED lamp works on the principle of converting electrical energy into
(a) Heat energy
☒ (b) Light energy
(c) Mechanical energy
(d) Sound energy. (Page 27)
- If a torch is not working even when the switch is turned on, which of the following could be the reason?
(a) The bulb is fused.
(b) The cells are not connected properly.
(c) The metal parts of the torch do not make proper contact with the cell.
☒ (d) All of the above. (Page 28)
- Sometimes the electric bulb does not glow even if it is connected to a cell by wires without any gap in the circuit. This can be occur due to
(a) broken bulb
☒ (b) improper wiring
(c) no flow of the current
(d) all of these. (Page 30)
- What is the function of a switch in a circuit?
(a) To increase the current.
(b) To decrease the voltage.
☒ (c) To control the flow of current.
(d) To provide energy to the circuit. (Page 32)
- Which of the following components is used to store electrical energy?
☒ (a) Battery (b) Switch
(c) Wire (d) Bulb (Page 32)

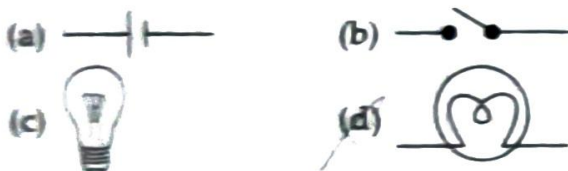
13. When a switch is in OFF position,
- (i) circuit starting from the positive terminal of the cell stops at the switch.
 - (ii) circuit is open.
 - (iii) no current flows through it.
 - (iv) current flows after some time.
- Choose the combination of correct answer from the following.
- (a) All are correct
 - (b) (ii) and (iii) are correct
 - (c) Only (iv) is correct
 - (d) Only (i) and (ii) are correct

(NCERT Exemplar)

14. Key or switch in a circuit is placed
- (a) left side of the battery
 - (b) right side of the battery
 - (c) anywhere in the circuit
 - (d) near the positive terminal of the bulb.

(Page 32)

15. Which of the following is the symbol of a bulb?

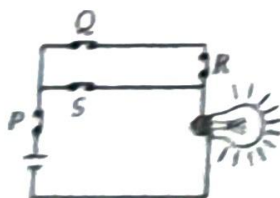


(Page 33)

16. The shorter, thicker vertical line in the symbol of a cell represents
- (a) the positive terminal
 - (b) the negative terminal
 - (c) the direction of current
 - (d) none of these.

(Page 33)

17. Which one of the following switch if opened, will cause the light bulb to stop glowing in the given circuit?



- (a) P
- (c) R

- (b) Q
- (d) S

(Page 34)

Effect of Electrical Current

18. The bulb in an electric circuit glows because of
- (a) magnetic effect of current
 - (b) heating effect of current
 - (c) chemical effect of current
 - (d) none of these.

19. An electric heater is used by us to convert electrical energy to

- (a) heat energy
- (b) light energy
- (c) chemical energy
- (d) both (a) and (b).

20. The device used to prevent the flow of excess current in the circuit is

- (a) switch
- (b) fuse
- (c) bulb
- (d) coil.

21. Mark out the correct statement.

- (a) Fuses and MCBs are safety device.
- (b) Fuses are safety device but MCBs are not.
- (c) MCBs are safety device but fuses are not.
- (d) Neither fuses nor MCBs are safety device.

22. The necessary requirement for a fuse wire is that, it should be a

- (a) conductor with low melting point
- (b) conductor with high melting point
- (c) insulator with low melting point
- (d) insulator with high melting point.

23. Which mark is necessary on electrical appliances?

- (a) ISI
- (b) FICC
- (c) AGMARK
- (d) KSK

24. The SI unit of charge is

- (a) Coulomb
- (b) Joule
- (c) Ampere
- (d) Watt.

25. Which of the following is not a circuit element?

- ✓(a) Potential difference
(b) Voltmeter
(c) Resistor
(d) Battery R
26. Magnetic effect of current was discovered by
(a) Thomas Jefferson
✓(b) Hans Christian Oersted
(c) Flemming
(d) Louis Pasteur R
27. If you place a compass near a current carrying wire, it will
✓(a) get deflected (b) get charged
(c) get hot (d) glow brightly. U

Electromagnet, Electrical Conductor and Insulator

28. Which of the following is not a source of electric current?
(a) Car battery (b) Dry cell
(c) Cell ✓(d) Electromagnet R
29. A current carrying coil of an insulated wire wrapped around a piece of soft iron becomes
✓(a) a filament ✓(b) an electromagnet
(c) a fuse (d) a conductor. R
30. Iron is used for making electromagnet, not steel, nickel, cobalt because
(a) iron is cheap and easily available.
(b) iron is a good conductor of electricity.
✓(c) when current is switched off in the coil of an electromagnet made of iron, iron loses all its magnetism quickly. U
(d) none of these.
31. Which of the following statements regarding an electromagnet is not true?
(a) Insulated wire is wound around it.
(b) It works only in the presence of electricity.
(c) It is used in fans and radio.
✓(d) It is a permanent magnet. U
32. The electric bell makes use of
(a) a horse-shoe shaped electromagnet
(b) a bar-shaped electromagnet
(c) a ring-shaped electromagnet
✓(d) disc shaped electromagnet. R
33. Which of the following is the most conductive element?
(a) Copper (b) Iron
(c) Silicon ✓(d) Silver
34. Why is copper a preferred choice in making electrical wires?
✓(a) Copper is a good conductor of electricity.
(b) Poor conductors of electricity.
(c) Insulators.
(d) Superconductors.
35. Conductors are materials that
✓(a) allow the flow of current
(b) does not allow heat to current
(c) stops current from passing through
(d) none of these.
36. Which one of the following is the best conductor of electricity?
✓(a) Silver (b) Copper
(c) Aluminium (d) Iron R
37. Why is electrical wiring usually made from copper and not silver?
(a) Copper is a better conductor.
(b) Copper is a better insulator.
✓(c) Copper is less expensive.
(d) Copper is non magnetic. U
38. Which of the following does not belong to the group formed by the others?
(a) Copper coin (b) Steel spoon
✓(c) Wooden ruler (d) Iron nail R
39. Why is electrical wiring usually covered with a layer of plastic?
(a) To make it strong.
(b) To help electricity flow in it.
✓(c) To make it safe. U
(d) To make it beautiful.

40. Which among the following is/are an insulator?
- (a) Wool (b) Plastic
(c) Silver (d) Both (a) and (b)

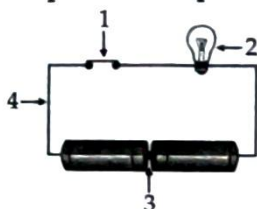
LEVEL - 2

41. In which circuit will the bulb or bulbs glow the brightest?
- (a) A simple circuit with one bulb and one cell.
(b) A simple circuit with one bulb and two cells.
(c) A simple circuit with two bulbs and one cell.
(d) A simple circuit with two bulbs and two cells.

42. Which one of the following phenomenon occurs when two wires of electricity supply line touch each other?
- (a) Overloading (b) Short circuiting
(c) Lightening (d) None of these.

43. Choose the statement which is not correct in the case of an electric fuse.
- (a) Fuses are inserted in electric circuits of all buildings.
(b) There is a maximum limit on the current which can safely flow through the electric circuits.
(c) There is a minimum limit on the current which can safely flow in the electric circuits.
(d) If a proper fuse is inserted in a circuit it will blow off if current exceeds the safe limit. (NCERT Exemplar)

44. The bulb in the given circuit does not glow. Which labelled part is responsible for this?



- (a) 1 (b) 2
(c) 3 (d) 4

45. Ravi has connected two bulbs across two cells in a simple circuit. How can he make the bulbs dimmer?
- (a) Replace one of the cells with a cork.
(b) Replace one of the cells with a wire.
(c) Replace one of the bulbs with a cork.
(d) Replace one of the bulbs with a wire.

46. The strength of an electromagnet does not depend upon
- (a) the current flowing through the coil
(b) the number of turns of the coil
(c) the insulating material of the coil
(d) the length of the electromagnet.

47. When the current flowing through a conductor is reversed, the deflection produced in a compass placed near it is
- (a) in same direction
(b) in opposite direction
(c) remains same
(d) none of these.

48. Read the given statements and select the correct option.

Statement 1 : An electric bulb works on the principle of heating effect of current.

Statement 2 : In a bulb, the filament is broken when a small current is passed through the bulb.

- (a) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.
(b) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.
(c) Statement 1 is true but statement 2 is false.
(d) Statement 1 is false but statement 2 is true.

49. In which of the following conditions, amount of heat produced in a current carrying wire, will increase?

- (a) When amount of current passing through the wire is increased.
(b) When amount of current passing in the wire is decreased.

- (c) When terminals of the wire are changed.
 (d) None of these.

An

50. How does the resistance of an ohmic conductor depend on the applied voltage?
 (a) Directly proportional.
 (b) Inversely proportional.
 (c) The resistance does not depend on applied voltage.
 (d) None of these.

R

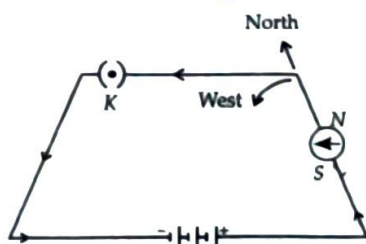
51. When one end of an iron rod is placed near a compass,
 (a) the N-pole of the compass needle always point towards it
 (b) the S-pole of the compass needle always point towards it
 (c) any pole of the compass needle may point towards it
 (d) the compass needle will not be affected by the iron rod.

R

52. Which coil produces the strongest electromagnet for a given flow of current?
 (a) A 10 cm coil with 100 turns.
 (b) A 5 cm coil with 200 turns.
 (c) A 10 cm coil with 200 turns.
 (d) A 20 cm coil with 200 turns.

Ap

53. A thick copper wire carries current due north placed parallel to and over a compass needle as shown in figure.

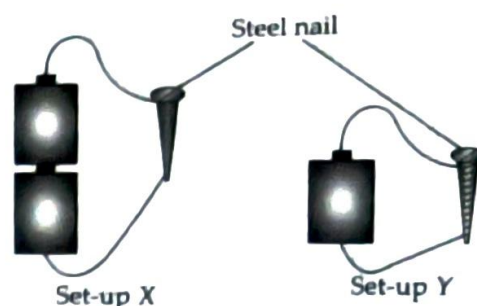


If the direction of current is changed from north to south, the needle will turn towards

- (a) west (b) north
 (c) south (d) east

An

54. Leena had set up the experiment as shown in the figure.



She found out that the steel nail in set-up X had attracted more iron pins than the steel nail in set-up Y. What could be the reason for this?

- I. The batteries used in set-up X are more expensive.
 II. There are more batteries used in set-up X.
 III. The number of turns of coil around the nail in set-up Y is fewer than in set-up X.
 IV. A longer wire was used in set-up X.
 (a) I and III only (b) II, III and IV only
 (c) II and III only (d) I, II, III and IV

An

55. The temperature of the filament in an incandescent lamp is critical to its output. What happens to the amount of visible light as the temperature increases?
 (a) It decreases.
 (b) It stays the same.
 (c) It increases.
 (d) It fluctuates randomly.

56. When connected in a circuit, the longer terminal of an LED is the
 (a) negative terminal
 (b) positive terminal
 (c) it does not matter which terminal is positive or negative.
 (d) this depends on the color of the LED.

57. Why does an LED lamp consume less power than an incandescent lamp?

- (a) It produces more heat.
 (b) It is designed to convert most of the energy into light.
 (c) It uses a different type of wire.
 (d) It is not connected to a power source.

58. In an electric circuit, what happens when the switch is in the 'OFF' position?
- The circuit is open, and no current flows.
 - The circuit is closed, and current flows.
 - The bulb lights up.
 - The battery gets charged.
59. Why are metals like copper used for electrical wires?
- They are good insulators.
 - They are good conductors of electricity.
 - They are poor conductors of electricity.
 - They are poor insulators.
60. Why are insulators important in an electrical circuit?
- They make the circuit safe to touch.
 - Electricity will only flow through insulators.
 - Light bulbs and buzzers are only made of insulating materials.
 - All of the above.
61. Choose the correct statement.
- A switch is the source of electric current in a circuit.
 - A switch help to complete or break the circuit.
 - A switch does not help us to use electricity as per our requirement.
 - When the switch is closed, there is an air gap between its terminals.
62. A current is passed through a steel wire heating it to red hot. In order to increase the temperature of the wire, one should
- increase the current passing through the wire.
 - decrease the current passing through the wire.
 - decreases the potential difference between the wire.
 - none of these.
63. When a magnetic compass needle is carried nearby to a straight wire carrying current,
- the straight wire causes a noticeable deflection in the compass needle.
 - the alignment of the needle is tangential to an imaginary circle with straight wire as its centre and has a plane perpendicular to the wire.
- (I) is correct.
 - (II) is correct.
 - Both (I) and (II) are correct.
 - Neither (I) nor (II) is correct.
64. The work done in moving a 10 C charge across two points is 350 J. What is the potential difference between these two points?
- 7 V
 - 35 V
 - 700 V
 - 3500 V
65. Assuming that the charge of an electron is 1.6×10^{-19} coulomb, the number of electrons passing through a section of wire per second when the wire carries a current of one ampere is
- 0.625×10^{19}
 - 1.6×10^{-19}
 - 1.6×10^{19}
 - 0.625×10^{-17}
66. A torch bulb which normally works on 4.5 V electricity is connected battery of 5 V. Which of the following will happen to this torch bulb?
- It will glow brightly.
 - It will glow very dimly.
 - It will not glow.
 - It will get fused.
67. Which of the following precautions need not be taken while using electric gadgets/appliances/circuit?
- We should never touch a lighted electric bulb connected to the mains.
 - We should never experiment with the electric supply from the mains or a generator or an inverter.
 - We should never use just any wire or strip of metal in place of a fuse.
 - We should never turn the switch in ON position.

Ans

Ap

Ap

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(NCERT Exemplar)

68. Current flows through a conductor connected across a voltage source. Now the resistance of the conductor is reduced to one fourth to its initial value and connected across the same voltage source. The heating effect in the conductor will become

- (a) half (b) double
(c) four times (d) one fourth. 

69. When more than one electric cell have to be used, they are joined to the electric circuit in such a way that

- (a) the positive terminal of one electric cell is connected to the positive terminal of another electric cell.
(b) the negative terminal of one electric cell is connected to the negative terminal of another electric cell.
(c) the positive terminal of one electric cell is connected to the negative terminal of another electric cell.
(d) both the cells are parallel to each other with similar terminals in same side.

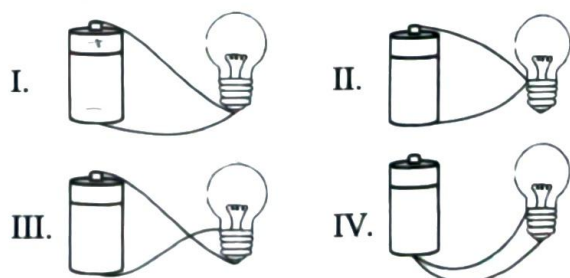
70. Read the following statements carefully and select the correct option.

- (i) In an electric cell, chemical energy is converted into electrical energy.
(ii) Combination of electric cells is called a battery.
(a) Only (i) is true.
(b) Only (ii) is true.
(c) Both (i) and (ii) are true.
(d) Both (i) and (ii) are false.

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

71. In which of the following cases the bulb(s) will glow?



- (a) I, II and IV only (b) I and IV only
(c) I and II only (d) III only

72. If the two terminals of a cell are connected directly with a wire, then

- (a) more electrical energy will be stored in the cell
(b) no current will flow in the wire
(c) the chemical gets used up very fast
(d) nothing will happen.



73. Which is the right way of joining cells?

- (a) 
(b) 
(c) 
(d) 

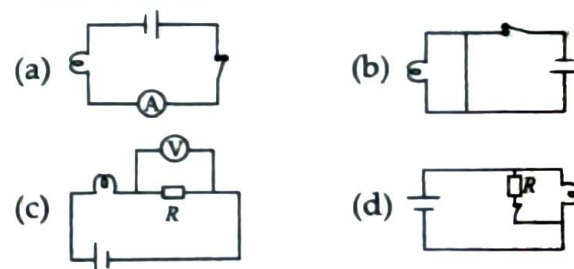
74. Why are the two terminals of the bulb not allowed to touch each other?

- (a) Electric sparks will occur.
(b) The bulb will start glowing.
(c) The bulb will not glow.
(d) More electricity will be used.

75. How do we check whether an electromagnet is working?

- I. Check if it can repel glass.
II. Check to see whether it conducts heat easily.
III. Place some paper clips near it.
(a) I only (b) III only
(c) I and II only (d) II and III only

76. Which of the following diagrams represents a short circuit?



77. Match column I with column II and select the correct option from the given codes.

Column I

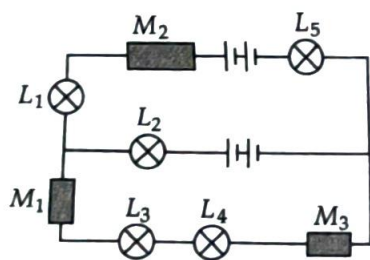
Column II

- | | |
|---|-----------------|
| P. The material of the filament of a bulb | (i) Tap water |
| Q. Electric source | (ii) Pure water |
| R. Conductor | (iii) Tungsten |
| S. Insulator | (iv) Dry cell |
- (a) P - (iv), Q - (iii), R - (i), S - (ii)
 (b) P - (iii), Q - (iv), R - (ii), S - (i)
 (c) P - (iii), Q - (iv), R - (i), S - (ii)
 (d) P - (iv), Q - (iii), R - (i), S - (ii)

78. Ramesh wants to increase the strength of his electromagnet. Which of the following can he do to make the electromagnet stronger?

- I. Increase the number of times the magnet is stroked.
 II. Increase the number of batteries.
 III. Increase the number of coils used around the electromagnet.
 IV. Change the thickness of the wire.
- (a) I and III only (b) II, III and IV only
 (c) II and III only (d) I, II and IV only

79. Five lamps L_1 , L_2 , L_3 , L_4 and L_5 and three different types of materials M_1 , M_2 and M_3 are used in the circuit shown.



If L_2 , L_3 and L_4 are the only lamps that light up, then which one of the following conclusions can be made?

- (a) Only M_2 and M_3 are insulators of electricity.
 (b) Only M_1 and M_3 are insulators of electricity.
 (c) Only M_1 and M_2 are conductors of electricity.
 (d) Only M_1 and M_3 are conductors of electricity.

80. Match column I with column II and select the correct option from the given codes.

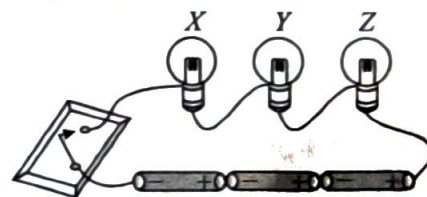
Column I

Column II

- | | |
|---------------------|---|
| P. Electric circuit | (i) It is a coil of insulated copper wire wrapped around an iron nail. |
| Q. Electromagnet | (ii) It is a combination of two or more cells. |
| R. Electric switch | (iii) It is an arrangement consisting of source of electricity, connectors and utilizers. |
| S. Battery | (iv) It is used in an electric circuit to make or break the circuit. |

- (a) P-(i), Q-(ii), R-(iv), S-(iii)
 (b) P-(iii), Q-(i), R-(iv), S-(ii)
 (c) P-(i), Q-(iii), R-(ii), S-(iv)
 (d) P-(iii), Q-(iv), R-(i), S-(ii)

81. Three bulbs X, Y and Z are connected in a circuit as shown in figure. When the switch is put on, then

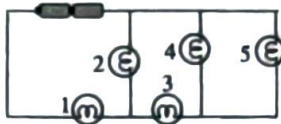


- (a) all the bulbs will glow at the same time
 (b) bulbs will glow in the order of X, Y and Z
 (c) bulb X will glow first whereas bulbs Y and Z will glow simultaneously after some time
 (d) bulb Z will glow first.

82. If 5 coulombs of charge flows through a conductor in 2 seconds, then the number of electrons flowing through a conductor in one second approximately is

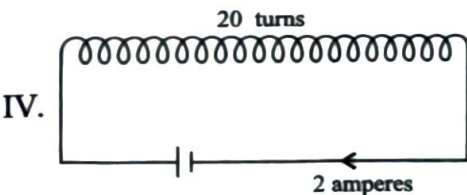
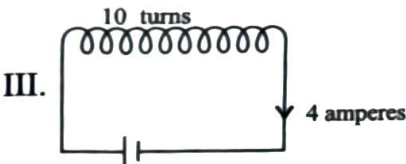
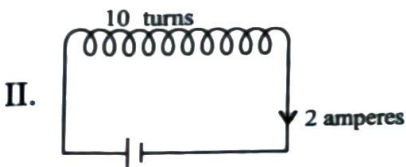
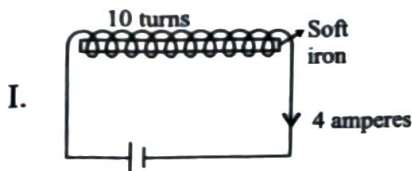
- (a) 30×10^{18} (b) 15×10^{18}
(c) 6×10^{18} (d) 12×10^{18}

83. The given diagram shows five bulbs connected to two dry cells. When one of the bulbs is faulty, the other four bulbs remain lighted. Which bulb is faulty?



- (a) 1 only (b) 3 only
(c) 2 or 4 only (d) 2 or 4 or 5 only

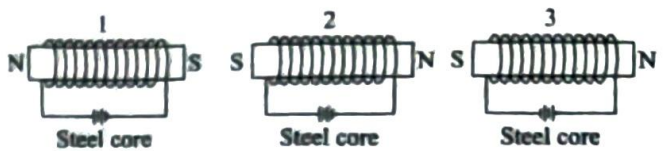
84. There are four electromagnets as shown in the figure.



Which one of following options is correct regarding strength of electromagnets?

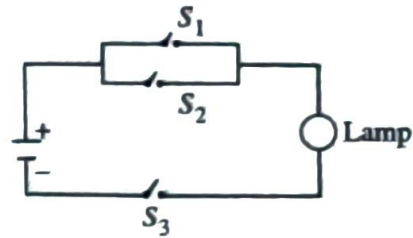
- (a) $I = II < III < IV$ (b) $I > II > III = IV$
(c) $I < II < III = IV$ (d) None of these

85. In which diagram(s) is/are the correct poles of the electromagnet indicated? (N indicates north pole and S indicates south pole.)



- (a) 2 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

86. In the given circuit diagram, S_1 , S_2 and S_3 are switches. Lamp will glow when

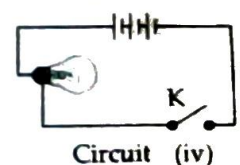
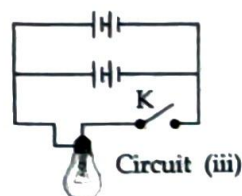
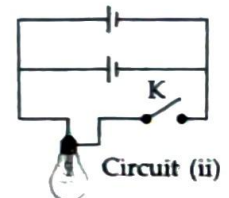
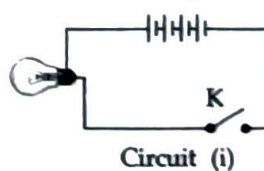


- | S_1 | S_2 | S_3 |
|------------|--------|--------|
| (a) open | open | closed |
| (b) closed | closed | open |
| (c) open | closed | open |
| (d) closed | open | closed |

87. Marking on an electric kettle is 500 W, 220 V. What does it signify?

- (a) When 220 V is applied across the kettle, 500 units of current flow in it.
(b) When 500 W is applied across the kettle, 200 units of current flow in it.
(c) When 220 V is applied across the kettle, 500 joules of energy is consumed for every second.
(d) When 500 W is applied across the kettle, 220 joules of energy is consumed for every second.

88. Study the four electric circuits (i), (ii), (iii) and (iv) carefully. [One cell (—|—) is equivalent to one volt.]



Similar bulbs were used in the four circuits. When the switches in the electric circuits were closed, the light bulbs lit up. Arrange the circuits, (i), (ii), (iii) and (iv) according to the brightness of their bulbs in ascending order.

- (a) (i), (ii), (iv), (iii) (b) (i), (iv), (iii), (ii)
(c) (ii), (iii), (iv), (i) (d) (iii), (i), (iv), (ii)

89. Mohit puts two different objects in each of the following containers.

Container (i) : copper coin and steel nail

Container (ii) : plastic toy and silver ring

Container (iii) : rubber and gold coin

Container (iv) : nickel coin and iron key

He can use a magnet to separate the objects in container(s)_____.

- (a) (i) only
(b) (i) and (iii)
(c) (i), (ii) and (iv)
(d) (i), (ii), (iii) and (iv).

90. Which of the following statements are incorrect?

- (i) Human body is a bad conductor of electricity.
(ii) The air gap surrounding an electric circuit acts as an insulator.
(iii) A battery always has two terminals.
(iv) When electricity lights up a bulb, no heat is produced.

- (a) (i) and (ii) only
(b) (iii) and (iv) only
(c) (i) and (iv) only
(d) (ii) and (iii) only

Fill in the Blanks

- The combination of two or more cells is called _____.
- The bulb will glow in a circuit only when the switch is in the _____ position.
- If the switch is in the _____ position, the bulb will not glow.
- The switch in a circuit is used to control whether the circuit is _____ or _____.
- An electric _____ is a complete pathway for the electric current to pass through.
- A thin coiled wire inside an electric lamp is a _____ that glows when electric current passes through it.
- The top metal tip of an electric cell is its _____ terminal.
- Longer line in the symbol for a cell represents its _____ terminal.
- When current is switched 'ON' in a room heater, it becomes _____ and _____.
- The safety device based on the heating effect of electric current is called a _____.
- In all the circuit diagrams, the direction of the current is shown by arrows, starting from the _____ of the cell or battery and ending in _____.
- A circuit may get _____ due to reasons like overloading.
- When an electric current flows through a wire, it behaves like a _____.
- The melting point of a fuse is _____ than the melting point of remaining circuit.
- Doctors use very small but strong _____ to remove iron dust from a patient's eye.
- If the amount of current passing through the coil of an electromagnet is decreased, the strength of the electromagnet will _____.
- A battery, made up of more than one cell is used to provide _____ current in the circuit, than a single cell.
- Electric current flows where there is some difference in _____.

True or False

1. An insulator does not allow the electric current to pass through it.
2. Electric current flows only when the cell is connected to a switch.
3. Electricity from an electric cell is dangerous if used with naked hands.
4. A lamp with a broken filament is fused lamp.
5. An electric switch in 'OFF' position eases the flow of electric current.
6. To make a battery of two cells, the negative terminal of one cell is connected to the negative terminal of the other cell.
7. When the electric current through the fuse exceeds a certain limit, the fuse wire melts and breaks.
8. An electromagnet does not attract a piece of iron.
9. An electric bell has an electromagnet.
10. An MCB can be used instead of a fuse in an electrical circuit.
11. To make a battery of two cells, the negative terminal of one cell is connected to the negative terminal of the other cell.
12. When the electric current through the fuse exceeds a certain limit, the fuse wire melts and breaks.
13. An electromagnet does not attract a piece of iron.
14. An electric bell has an electromagnet.
15. Insulators do not carry electric current because they do not have free electrons.

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as

options (a), (b), (c) and (d) out of which one is correct.

1. List-I (Component)	List-II (Function)
(P) Battery	(1) Conducts electricity between components
(Q) Switch	(2) Lights up when current flows through it
(R) Bulb	(3) Controls the flow of current in the circuit
(S) Wire	(4) Provides electrical energy to the circuit

- (a) P-4, Q-3, R-2, S-1
 (b) P-3, Q-4, R-1, S-2
 (c) P-4, Q-2, R-3, S-1
 (d) P-3, Q-4, R-1, S-2

2. List-I	List-II
(P) LED or incandescent	(1) An electric circuit
(Q) An electric circuit diagram	(2) Electric cell
(R) The path provided to electricity to flow through an electric wire or a device	(3) Plastic or rubber
(S) A portable source of electric current	(4) Electric lamp
(T) Conducting wires are covered with	(5) Uses symbols

- (a) P-4, Q-1, R-5, S-2, T-3
 (b) P-5, Q-4, R-3, S-1, T-2
 (c) P-5, Q-4, R-3, S-2, T-1
 (d) P-4, Q-5, R-1, S-2, T-3

3. **List-I** **List-II**
- | | |
|-----------------|----|
| (P) Bulb | 1. |
| (Q) Cell | 2. |
| (R) Battery | 3. |
| (S) Wire | 4. |
| (T) Open switch | 5. |
- (a) P-2, Q-3, R-5, S-1, T-4
 (b) P-5, Q-4, R-2, S-1, T-3
 (c) P-5, Q-2, R-4, S-1, T-3
 (d) P-1, Q-2, R-3, S-4, T-5

4. **List-I** **List-II**
- | | |
|---------------------|--|
| (P) CFL | 1. A safety device to prevent damage due to short circuit. |
| (Q) ISI Mark | 2. The wire used in heater to get heat. |
| (R) Fuse | 3. Consume less energy than bulb. |
| (S) Heating element | 4. Certifies the safety of electrical appliances. |
- (a) P-1, Q-2, R-3, S-4
 (b) P-3, Q-4, R-2, S-1
 (c) P-3, Q-4, R-1, S-2
 (d) P-2, Q-1, R-4, S-3

Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (c) If assertion is true but reason is false.
 (d) If assertion is false and reason is true.

1. **Assertion :** A battery is used to power an electrical circuit.
Reason : A battery provides a source of voltage to move electrons through the circuit.

2. **Assertion :** The switch in a circuit is used to control the flow of electricity.
Reason : The switch opens or closes the circuit to either allow or stop the flow of current.

3. **Assertion :** An electric switch is not a necessary component of electric circuits so it must be avoided in circuits to save electricity.
Reason : An electric switch is used to make or break the circuit when required.

4. **Assertion :** Fuses are used in electric circuits.
Reason : Fuse prevents electric fires and damage to electrical appliances due to excessive flow of current.

5. **Assertion :** Copper wires are used for making connections in a circuit as they are good conductors of electric current.
Reason : When current passes through the copper wire, it behaves like a magnet.

6. **Assertion :** The amount of heat produced in a wire depends on its material.
Reason : If a small current passes through the filament of an electric bulb, the filament may become so hot that it may even melt and break.

7. **Assertion :** The strength of an electromagnet can be increased by increasing the number of turns of the coil per unit length.
Reason : A soft iron piece, which is kept in a coil of wire, behaves like a magnet when electricity is passed through the wire.

8. **Assertion :** Soft iron is used for making electromagnets.
Reason : Iron piece loses all its magnetism quickly when current is switched off.

Comprehension Type

PASSAGE-I : An electric circuit is a continuous and closed path through which electric current flows. While representing an electric circuit, some common electrical components can be represented by using standard symbols. Look at the given symbol carefully and answer the following questions.



- It is a symbol of
 - switch
 - fuse
 - bulb
 - cell.
- The figure is made up of two parallel lines. One is thin and long, while the other is thick and short. The thin and long line represents
 - negative terminal of the cell
 - positive terminal of the cell
 - negative terminal of the bulb
 - positive terminal of the bulb.

PASSAGE-II : Electric heater, electric iron, water heater etc., work on the principle of heating effect of current. When these electric appliances are connected to the main supply of electricity, become hot but the connecting wire remain cold.

- The element of electric heater is made of
 - nichrome
 - aluminium
 - lead
 - tungsten.
- Which of the following term represent heat energy?
 - VI
 - I^2Rt
 - IR^2
 - V^2/R

PASSAGE-III : Those substances through which electric charges cannot flow are called insulators. In other words, Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, bakelite, porcelain, and dry air, are all insulators.

In some substances, the electric charge can flow easily while in others they cannot. Those substances through which electric charge can flow, are called conductors. All metals and metal alloys such as nichrome, manganin and constantan are also conductors.

- If we touch a naked current carrying wire, we get a shock. This is because our body is a
 - conductor of electricity
 - insulator of electricity
 - source of electricity
 - both (b) and (c).
- Which of these conducts electricity?
 - Leather jacket
 - Plastic mug
 - Safety pin
 - Eraser
- Switches of any circuit are made up of plastics because they are
 - conductors
 - insulators
 - reflectors
 - easy to use.

Subjective Problems

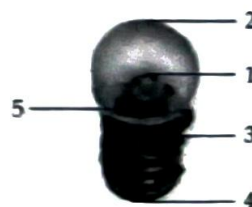
Very Short Answer Type

- How can we connect the cells to prepare a battery?
- Does the bulb glow when the circuit is open?
- A fused bulb does not glow. Why?
(NCERT Exemplar)
- In which of the following circuits A, B and C given in figure, the cell will be used up very rapidly?



(NCERT Exemplar)

- Figure shows a bulb with its different parts marked as 1, 2, 3, 4 and 5. Which of them label the terminals of the bulb?



(NCERT Exemplar)

6. Is it possible for electric current to flow in a bulb with a broken filament?
7. What is the meaning of the term 'battery'?
8. Name two sources of electricity.
9. What is the source of electric current in a torch?
10. Name the element from which the filament of electric bulb is made.
11. What do you mean by an electric circuit?
12. In which position is key placed in a circuit?
13. How can we connect the cells to prepare battery?
14. Name any four appliances which work on the principle of heating effect of current.
15. What are MCBs?
16. What type of magnet is used in an electric bell?
17. Name any four devices where magnetic effect of electric current is used.
18. What is resistance?
19. What is 'element' in respect of heating effect?
20. What are the effects of electricity?
8. What are the factors on which heating effect of electric current depends?
9. When does the overloading of a household electric circuit occur?
10. What are electric fuses? What is the importance?
11. Why does a magnetic needle get deflected when it is placed near a current carrying wire?
12. What happens when a compass needle brought near to a current carrying wire?
13. How are electromagnets prepared?
14. Draw the diagram of a simple electromagnet which works on the principle of magnetic effect of electric current.
15. What is the difference between a conductor and an insulator? Provide examples of each.

Long Answer Type

1. How does an electric torch work?
2. Explain the construction and working of an incandescent lamp.
3. What do you mean by an electric circuit? What are the main components of an electric circuit? Draw diagram of an electric circuit.
4. What is the function of a cell holder and a lamp holder in an electric circuit?
5. How can you show that an electric current has magnetic field?
6. What is an electromagnet? Name some devices in which it is used.
7. Explain the working of electric bell.
8. How will you show that an object is an electric conductor or an insulator?

Short Answer Type

1. Why should we not touch the lighted bulb?
2. Explain the construction of a battery.
3. What is the difference between an incandescent lamp and an LED lamp?
4. Why are metals like copper used for making electric wires?
5. Explain how an electric lamp can be made to glow using a cell or battery.
6. Explain the working of an electric switch in a circuit.
7. Why does an electric bulb get hot if it is switched on for a while?

Integer/Numerical Value Type

1. If 600 C of electric charge flows through a circuit, what will be the current (in A) drawn by the filament of an electric bulb in 10 minutes?

- The work done in moving a 20 C charge across two points is 400 J. The potential difference between these two points is x V. Find the value of x .
- A bulb draws a current of 0.22 A from a 220 V source. The resistance of the filament of the bulb is $x \Omega$. What is the value of x ?
- What happens when an electric cell is connected to a circuit?
 - The circuit melts
 - current flows through the circuit
 - The cell disappears
 - Nothing happens
- Which terminal of the cell does the electric current flow from?
 - Positive terminal
 - Negative terminal
 - Both terminal simultaneously
 - Middle of the cell
- What is the function of the positive and negative terminals in an electric cell?
 - To produce light
 - To store energy
 - To create flow of current
 - To increase voltage

Case Based Questions

Case I : A torchlight is a portable light source that runs on batteries. It uses electricity from the batteries to power an LED or an incandescent lamp, providing light in places without electrical outlets.

- What type of electricity powers a torchlight?
 - Electricity from a battery
 - Lightning
 - Electricity from the wall
 - Static electricity
- What is the function of a torchlight's LED or incandescent lamp?
 - To store energy
 - To produce light
 - To conduct electricity
 - To cool the battery
- What is the main component in a torchlight that stores energy?

(a) Switch	(b) Lamp
(c) battery	(d) Wire
- Which component of a torchlight is powered by the battery to provide light?

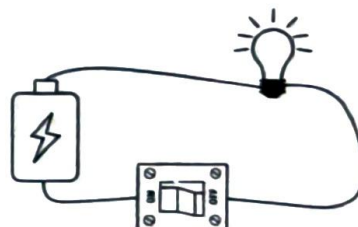
(a) Bulb or LED	(b) Switch
(c) Casing	(d) Terminal

Case II : An electric cell is a portable source of electrical energy. The positive terminal of the electric cell is marked with a + sign, and the negative terminal is marked with a – sign.

- What is the terminal marked with a “–” sign called?

(a) Neutral terminal	(b) Negative terminal
(c) Positive terminal	(d) Live terminal

Case III : On the basis of this experimental observations, a German physicist George Simon Ohm derived a relationship between electric current and potential difference in 1828. The current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the temperature and other physical conditions remain unchanged.
 Potential difference $\propto$ Current $\therefore V \propto I$ or $V = RI$
 The proportionality constant R is called the resistance of the conductor.



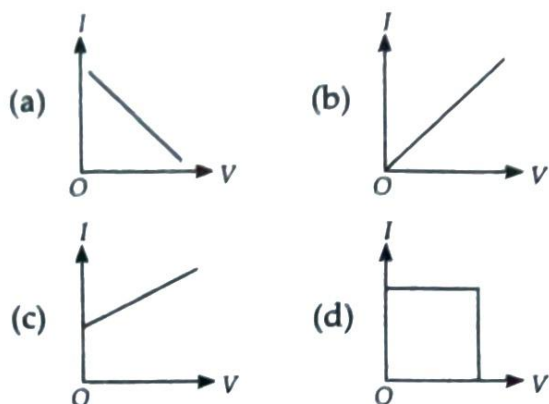
The resistance of a conductor is the property by virtue of which it opposes the flow of charges through it.

CFQs

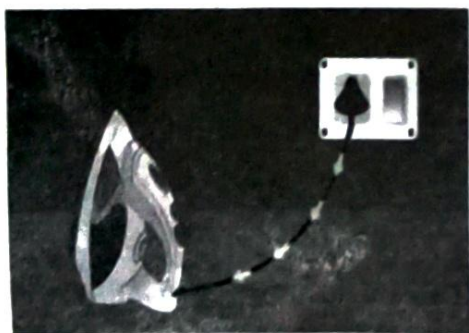
- The SI unit of resistance is

(a) mho	(b) ohm
(c) ampere	(d) voltmeter.

- The potential difference required across a conductor of resistance $5\ \Omega$ to make a current of 1.5 A flow through it is
(a) 7.5 V (b) 75 V (c) 0.75 V (d) 750 V
- In circuit, current becomes half when resistance is
(a) removed (b) doubled
(c) halved (d) it is in circuit
- When a 4 V battery is connected across an unknown resistor, there is a current of 100 mA in the circuit. The value of the resistance of the resistor is
(a) $4\ \Omega$ (b) $40\ \Omega$ (c) $400\ \Omega$ (d) $0.4\ \Omega$
- Choose the correct graph between V and I .
- If the current passing through a heater is halved, then the heat produced by it becomes
(a) $H/2$ (b) $4H$
(c) $H/4$ (d) $2H$
- How much heat energy is produced in 10 seconds in an electric iron having resistance $440\ \Omega$, when a potential difference of 220 V is applied across it?
(a) 1200 J (b) 600 J
(c) 1000 J (d) 1100 J
- Which property of the electricity is responsible for use of fuse wire in household wiring?
(a) Chemical effect (b) Heating effect
(c) Magnetic effect (d) All of above
- Find the potential difference across the resistor, when 125 J of heat is produced each second in a 5 ohm resistor.
(a) 25 V (b) 30 V
(c) 20 V (d) 28 V
- A bulb is rated as 270 V , 0.5 A . Its power is
(a) 125 watt
(b) 100 watt
(c) 60 watt
(d) 135 watt



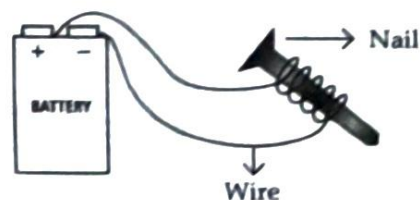
Case IV : Whenever a current is passed through a conductor, it becomes hot after sometime. This means that electrical energy is being converted into heat energy. Electric bulb, electric iron, electric heater etc., are some examples of appliances which are based on this effect, called the heating effect of current or the Joule heating.



Joule's law of heating : $H = I^2RT$

CFQs

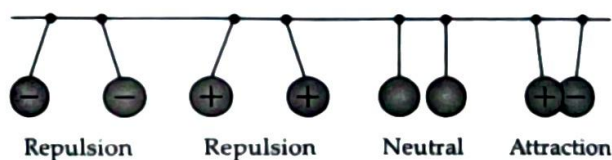
Case V : Electromagnet is a temporary strong magnet and is just a solenoid with its winding on a soft iron core. In order to provide a strong magnetic field in a small region, an electromagnet is made in the U-shape, such magnet is called horse-shoe magnet. An electromagnet produces a strong magnetic field whose strength can be changed by changing the current through the solenoid. The polarity of an electromagnet can easily be reversed by changing the direction of current through the solenoid.



CFQs

- The most suitable material for making the core of an electromagnet is
 - steel
 - iron
 - soft iron
 - aluminium
- Which of the following is not attracted by magnet?
 - Iron
 - Cobalt
 - Brass
 - Nickel
- The strength of magnetic field inside a long current carrying straight solenoid is
 - more at the ends than at the centre
 - minimum in the middle
 - same at all points
 - found the increase from one end to the other.
- The direction of magnetic field lines inside magnet is
 - North to South pole
 - South to North pole
 - there is no direction inside a magnet
 - can't be determine.
- Electromagnets used in
 - removing pieces of iron from wounds
 - electric fan
 - generator
 - all of the above.

Case VI :



Conductors contain a large number of free electrons which make them good conductor of electricity. When a charge body is brought in contact with the Earth, its entire charge passes to the ground in the form of a momentary current. It is found experimentally that the electric charge of any body, large or small is always an

integral multiple of a certain minimum amount of charge.

CFQs

- How many electronic charges form five coulomb of charge?
 - 1.6×10^{-19}
 - 3.125×10^{18}
 - 3.125×10^{19}
 - 1.6×10^{19}
- The charge carried by 12.5×10^8 electrons is
 - $2 \times 10^{-10} \text{ C}$
 - $2 \times 10^{10} \text{ C}$
 - $2 \times 10^{11} \text{ C}$
 - $2 \times 10^{-11} \text{ C}$
- If a comb drawn through person's hair on a dry day causes 10^{22} electrons to leave the person's hair and stick to the comb, then the charge carried by the comb is
 - $1.6 \times 10^{-22} \text{ C}$
 - $1.6 \times 10^{22} \text{ C}$
 - $1.6 \times 10^{-3} \text{ C}$
 - $1.6 \times 10^3 \text{ C}$
- A glass rod 'P' is observed to be attracting an ebonite rod 'Q', then
 - 'P' and 'Q' are uncharged bodies
 - both 'P' and 'Q' are positively charged
 - both 'P' and 'Q' are negatively charged
 - either 'P' is charged or 'Q' is charged. If both are charged, they are opposite in nature.
- A negatively charged body is connected to the ground, then choose the correct statement.
 - Electrons flow from body to ground.
 - It becomes positively charged.

- Only (A) is true.
- Only (B) is true.
- Both (A) and (B) are true.
- Both (A) and (B) are false.

Case VII : Consider a point charge $+q$ located at point O. Let A and B be two points in its electric field. When a test charge q_0 is moved from A to B, a work W has to be done in moving against

the repulsive force exerted by the charge $+q$. The potential difference between points A and B is

$$V = V_B - V_A = \frac{W_{AB}}{q_0} = \frac{\text{Work}}{\text{Charge}}$$



1. 5 J of work is done in moving a positive charge of 0.5 C between two points. The potential difference between these two points is

(a) 25 V (b) 2.5 V
(c) 10 V (d) zero

2. The SI unit of electric potential is
(a) ampere (b) volt
(c) joule (d) coulomb.

3. A charge of 2 coulomb moves between two points maintained at a potential difference of 1 volt. The energy acquired by the charge is

(a) 0.1 J (b) 0.2 J
(c) 1 J (d) 2 J

4. How much is the electric potential of a charge at infinity?

(a) Zero (b) Infinite
(c) One
(d) Can't determine

Heat Transfer in Nature

CHAPTER

02

Learning Outcomes...

- Heat and Temperature
- Thermometers
- Transfer of Heat
- Water Cycle

Introduction

We often use the word 'hot' and 'cold' while describing things around us. Suppose you have a bowl of hot soup or a cup of coffee that is too hot. What can you do to cool it? You would probably add something cooler such as ice.

When we mix ice to hot coffee it becomes cooler and ice starts melting. Something is getting transferred between the ice and hot coffee. **This something is actually a kind of energy which transfers from a hot body to a cold body and is known as heat.** This chapter is an attempt to elaborate different aspects of heat, temperature, their measurement and their effect.

Heat and Temperature

Heat is a form of energy which flows from a body at a higher temperature to lower temperature. Temperature is a physical quantity that measures the degree of hotness or coldness of a body. Heat can increase or decrease the temperature. The direction of flow of heat between two bodies is determined by their temperature.

DO YOU KNOW?

Chemical energy stored in fuels or food can be changed into heat energy. For example, burning wood, coal or candle releases heat and even the food we eat gives energy that our body partly use as heat.

ACTIVITY CORNER



To understand how heat flows from a body at a higher temperature to one at a lower temperature.

- ▶▶ Take some hot water in a tumbler. Dip a metal spoon in this water for some time. How long does it take for the spoon to become hot for you to feel uncomfortable to hold it anymore?
- ▶▶ This shows that heat is transferred from the hot water in the tumbler to the spoon and then to your hand.
- ▶▶ Leave the tumbler with water in it for 10-15 minutes. Feel the water now. Has it cooled down? Why?
- ▶▶ Water has lost its heat to the surroundings. This shows that heat has been transferred from the water to the surroundings.

